

Supplemental Material for: Non-additive Strain–Dopant Energetics in Quasi-Hexagonal C₆₀ Graphullerene

Xingqiang Chen and Qixing Wang
Department of Physics, Xiamen University, Xiamen 361005, China

I. LITERATURE POSITIONING AND DESCRIPTOR COMPARISON

II. ARCHIVED LEGACY-PBE TETRAMER BASELINE

Table III reproduces the archived rigid-strain grid (PBE without D3) retained for reproducibility; it is *not* used for quantitative claims in the main text.

III. CALCULATION INVENTORY (REPRODUCIBILITY)

Tables IV and V list the DFT groups underlying the main-text claims and superseded exploratory grids retained only for reproducibility. They are *not* scientific results tables.

IV. FULL TETRAMER π -DOS AT $\epsilon=0$

V. PERIODIC FRONTIER-ORBITAL LOCALIZATION

Cage-resolved VBM/CBM isosurfaces on periodic $n \times C_{60}$ cells ($n = 2-8$) at $\epsilon=0$ complement the tetramer π -DOS above and the main-text electronic taxonomy (donor/acceptor vs. size dilution). Carbon cages are shown as uniform gray MSMS molecular surfaces (probe 1.3 Å); heteroatoms use element colors.

VI. PERIODIC COUPLING-ENERGY ROWS ($n \geq 6$)

VII. MARCUS REORGANIZATION ENERGIES

Polaron transport diagnostics are reported here for completeness; the main-text analysis centers on S .

VIII. HOPPING INTEGRAL J

On the tetramer dimer grid, B substitution at $\epsilon=0$ delocalizes the charged-state orbital (IPR 75.0 $\rightarrow$ 24.6) while raising J from 26.6 to 46.4 meV; coupled B @ +3% gives $J = 37.2$ meV and IPR 23.1. The criterion $J > \lambda/2$ for polaron hopping dominance [4] is *not* met at PBE+D3

TABLE I. Positioning relative to representative prior work on qHP graphullerene and related covalent networks (not an exhaustive literature map). “Joint (ϵ, δ)” denotes simultaneous strain and substitutional doping in the same calculation series; “four-corner total-energy audit” denotes symmetric (ϵ, δ) corner differences as in the main-text definition of S .

Prior focus	Joint (ϵ, δ)	Four-corner
Separate strain or doping maps [1–5]	×	×
Nonlinear band-edge / transport response [6–8]	Partial	×
Alloy cluster expansion (effective $c\epsilon x$) [9]	✓	×
This work (qHP C ₆₀ , PBE+D3)	✓	✓

($J < \lambda/2$), so a polaron transition is not confirmed on this grid.

-
- [1] H. Li, M. A. Shakoory, and R. Wang, strain engineering of electronic structure and thermoelectric properties of quasi-hexagonal fullerene monolayer, *J. Appl. Phys.* **136**, 015101 (2024).
 - [2] M. A. Khan and M. N. Leuenberger, tuning the electronic and optical properties of impurity-engineered two-dimensional graphullerene half-semiconductors, *Phys. Rev. Mater.* **9**, 034001 (2025).
 - [3] B. Peng and M. Pizzochero, monolayer C₆₀ networks: a first-principles perspective, *Chem. Commun.* **61**, 10287 (2025).
 - [4] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa, electron localization and mobility in monolayer fullerene networks, *Nano Lett.* **24**, 7891 (2024).
 - [5] R. M. Tromer, L. A. R. Junior, and D. S. Galvao, A DFT study of the electronic, optical, and mechanical properties of a recently synthesized monolayer fullerene network, *Chem. Phys. Lett.* **804**, 139925 (2022).
 - [6] S. Chen, H. Liu, and K. Wang, Untangling the respective effects of heteroatom doping and strain on electronic transport in 2D materials, *Adv. Mater.* **36**, 2301234 (2024).
 - [7] P. Silva, R. M. Faria, L. G. Ferreira, and J. L. F. Fatima, large polarons in two-dimensional fullerene networks: the crucial role of anisotropy in charge transport, *Nanoscale* **16**, 2337 (2024).
 - [8] P. Santra, S. Ghaderzadeh, M. Ghorbani-Asl, H.-P. Komsa, E. Besley, and A. V. Krashenninnikov, strain-modulated defect engineering of two-dimensional materials, *npj 2D Materials and Applications* **8**, 33 (2024).

TABLE II. Positioning of the strain-dopant coupling energy $\mathcal{S}$ (main-text four-corner estimator) against common stability and electronic descriptors. Algebraically $\mathcal{S}$ has the form of an interaction (coupling) energy; the distinction is *dimensional*, i.e. which axes of the joint (ϵ, δ) control space each quantity resolves. “Additive-screening error” denotes the bias incurred when total energies under joint load are assembled additively from separate strain-only and doping-only scans; $\mathcal{S}$ equals that bias by construction. “Double-reference cross term” denotes a symmetric difference of differences over the four (ϵ, δ) corners rather than subtraction of a single reference state or fragment pair. Entries are conceptual scope, not numerical performance; consistent with its energetic definition, $|\mathcal{S}|$ is *not* reducible to a single geometric ($\Delta\bar{d}$) or charge (Δq) channel alone.

Descriptor	Nonlinear (ϵ, δ) coupling	Additive-screening error	Double-reference cross term	Joint strain + doping design
Formation / substitution energy E_{sub}	×	×	×	×
Binding / interaction energy ΔE_{int}	Partial	×	×	Partial
Linear strain coefficient α	×	×	×	Partial
Band gap E_g / work function	×	×	×	×
$\mathcal{S}$ (this work)	✓	✓	✓	✓

TABLE III. Archived legacy-PBE (no D3) tetramer rigid-strain baseline retained for reproducibility only; not used for main-text quantitative claims. $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ at $\epsilon=0$ (no chemical potentials; relative ranking within this protocol only).

System	n_{dop}	$E(\epsilon=0)$ (E_{h})	$E(\epsilon=+5\%)$ (E_{h})	α (meV/%)	E_{sub} (eV/dopant)
Pristine	—	−1363.5512	−1363.5509	−1.0	—
B-doped	13	−1328.4900	−1328.4090	+56.9	+73.4
N-doped	12	−1414.3812	−1414.4053	−303.2	−115.3
P-doped	12	−1370.4965	−1370.4888	−19.1	−15.8

[9] A. van de Walle and V. Blum, Cluster expansions in alloy theory, MRS Bull. **34**, 681 (2009).

TABLE IV. DFT calculation groups supporting the main-text claims (reproducibility inventory; Supplemental Material only). Superseded exploratory grids appear in Table V.

Calculation group	System	Count	Protocol
Archived baseline (legacy PBE)	$4 \times \text{C}_{60}$	24	Rigid strain; not used for main-text claims
π -DOS and gaps	$4 \times \text{C}_{60}$	12	PBE+D3
IPR (polaron dimer)	Dimer	4	Pristine/B; $\epsilon=0, +3\%$
Hopping J	Dimer	4	PBE+D3
Charged polarons ^a	$4 \times \text{C}_{60}$	20	UKS; geometry opt. + vertical SP
Periodic size scaling	$n \times \text{C}_{60}$	41	PBE+D3; 400/350 Ry
Cutoff benchmark	$6 \times \text{C}_{60}$ N	1	+3%; 400 Ry single-point
Ionic relaxation	$4 \times \text{C}_{60}$	4	Fixed-cell GEO; P corners
Ionic relaxation	$1 \times \text{C}_{60}$ P	4	Fixed-cell GEO; periodic
Alternate placement	$4 \times \text{C}_{60}$	24	Rigid PBE+D3
Periodic placement	$4 \times \text{C}_{60}$	16	$n=4$; two independent maps
Matched-functional rigid	$4 \times \text{C}_{60}$	4	PBE+D3 at relaxed coordinates
Reference placement	$4 \times \text{C}_{60}$	24	Rigid PBE+D3
Hirshfeld population	$1 \times \text{C}_{60}$ B/N/P	18	Spin-polarized KS

^aSix relaxed tetramer geometries provide charged-state initial structures.

TABLE V. Exploratory DFT grids superseded by the production protocol (reproducibility archive only).

Task group	System	N_{conv}	Disposition
Early strain probe	$2 \times \text{C}_{60}$	5	Superseded
Single-molecule doping	C_{60}	11	Superseded
Early electronics	$2 \times \text{C}_{60}$	24	Superseded
Exploratory design grid	$4 \times \text{C}_{60}$	20	Archive only

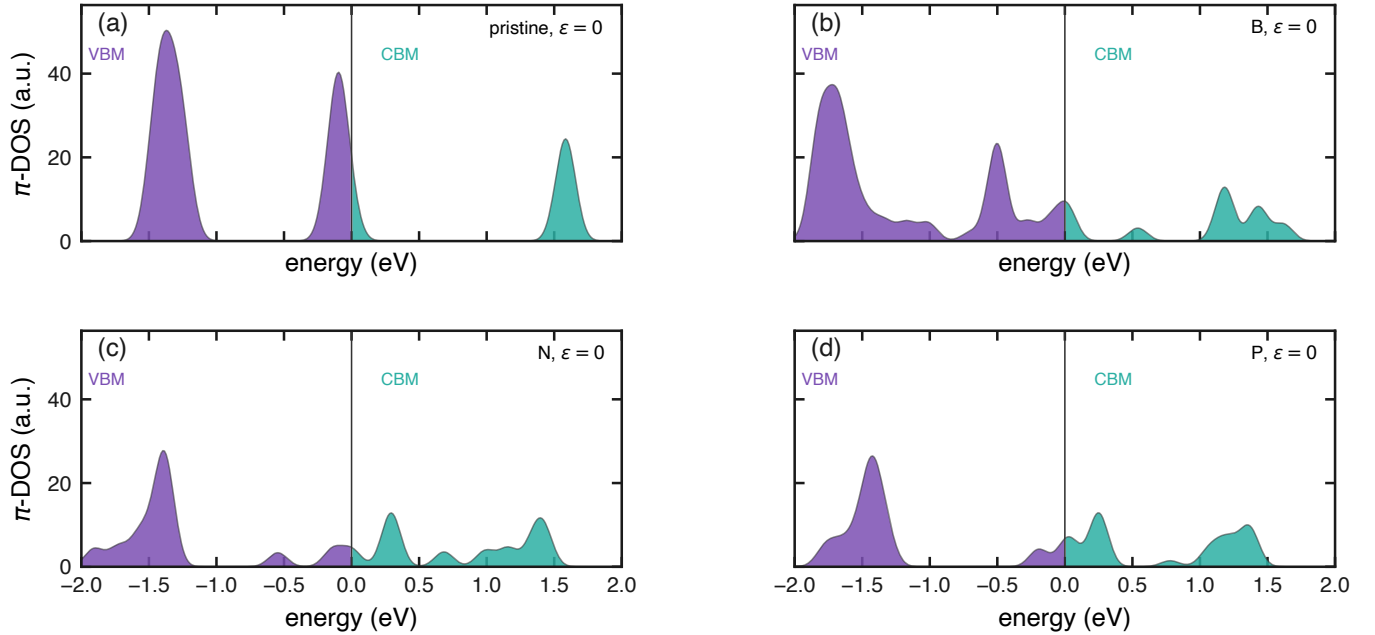
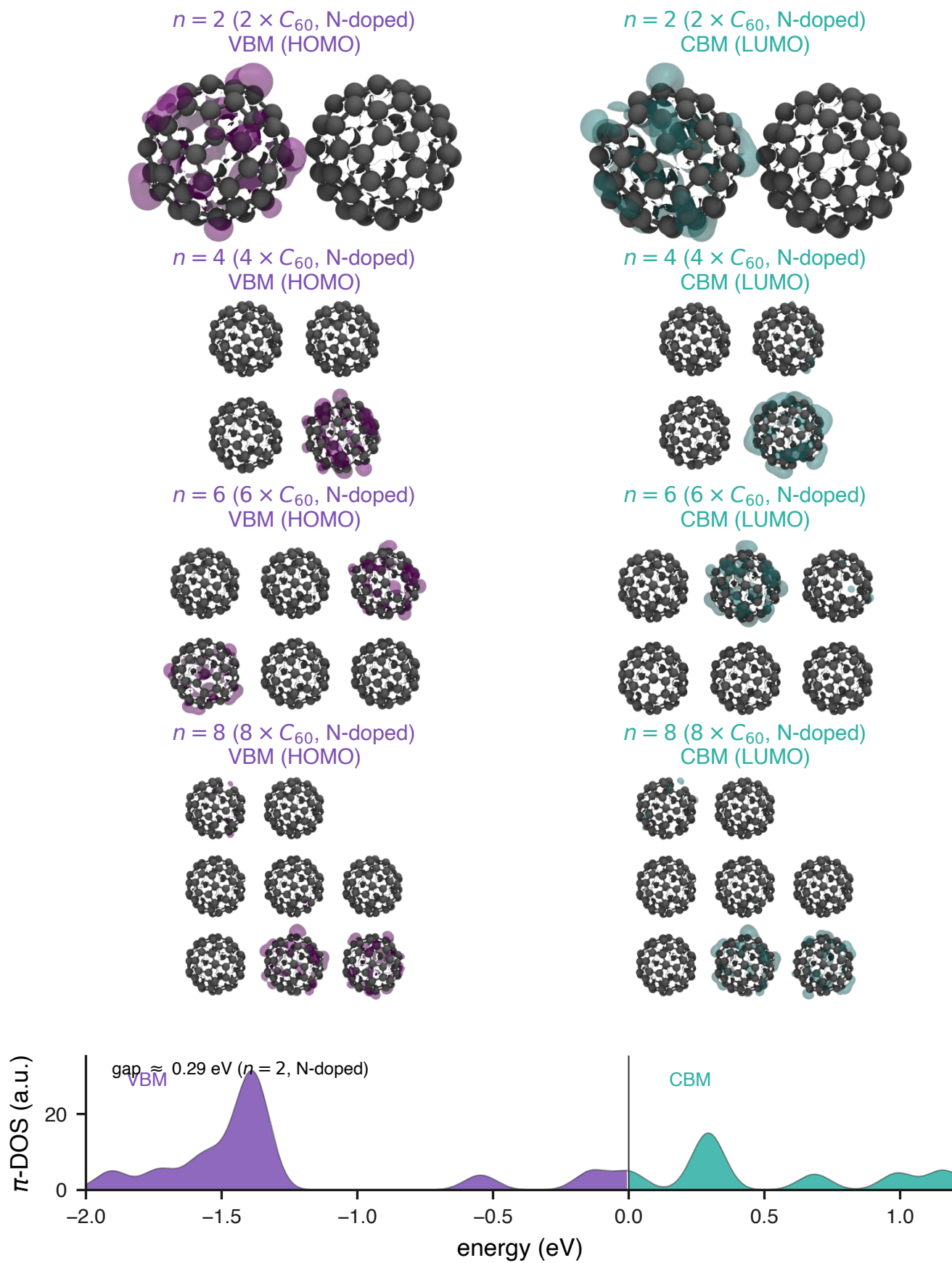


FIG. 1. Full π -DOS panels for pristine and B/N/P-doped tetramers at $\epsilon=0$ (PBE+D3), complementary to the main-text electronic-structure figure.

qHP C_{60} graphullerene ($\varepsilon = 0\%$, N-doped)

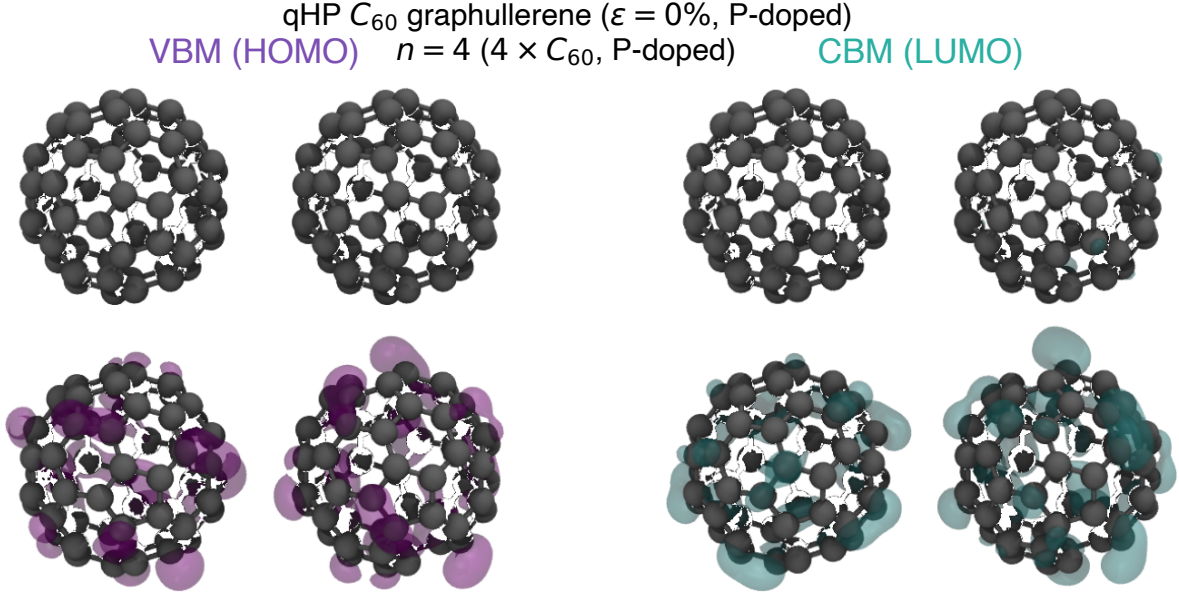
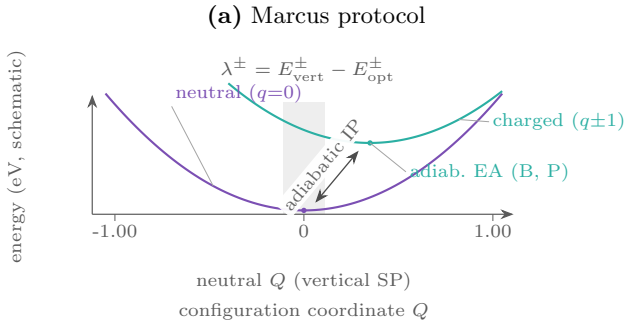


FIG. 3. P-doped $n=4$ supercell at $\epsilon=0$: VBM (left) and CBM (right) on gray MSMS cages. Both frontier states localize on a subset of cages, providing a spatial counterpart to the large periodic $|S|$ of size-mismatched P reported in the main text.

TABLE VI. Supplemental periodic coupling-energy rows at +3% ($n \geq 6$; 350 Ry production cutoff; completeness only).

n	Dopant	ϵ (%)	$\mathcal{S}$ (meV/atom)
6	B	+3	-7.76
6	N	+3	+2.38
6	P	+3	-31.89
8	B	+3	-11.26
8	N	+3	-2.30
8	P	+3	-29.94



(b) Adiabatic IP/EA and $\lambda^{\pm}$

Dopant	IP (eV)	EA (eV)	λ^{+} (eV)	λ^{-} (eV)
Pristine	4.73	2.61	0.041	0.042
N	3.89	2.77	0.058	0.065
B	4.11	3.12	0.051	— [†]
P	4.07	2.96	0.200	0.315

[†]B channel: $\lambda^{-} < 0$ under fixed neutral-geometry vertical SP; not reported.

FIG. 4. Marcus reorganization energies from charged-state geometry optimizations and vertical single-points at the neutral geometry. Panel (a) schematically defines the vertical-adiabatic decomposition following Ref. 4; panel (b) tabulates adiabatic IP/EA and $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$ for pristine and B/N/P tetramers. The electron-channel λ^{-} for B is unphysical under the fixed neutral-geometry vertical protocol and is excluded (Limitations).

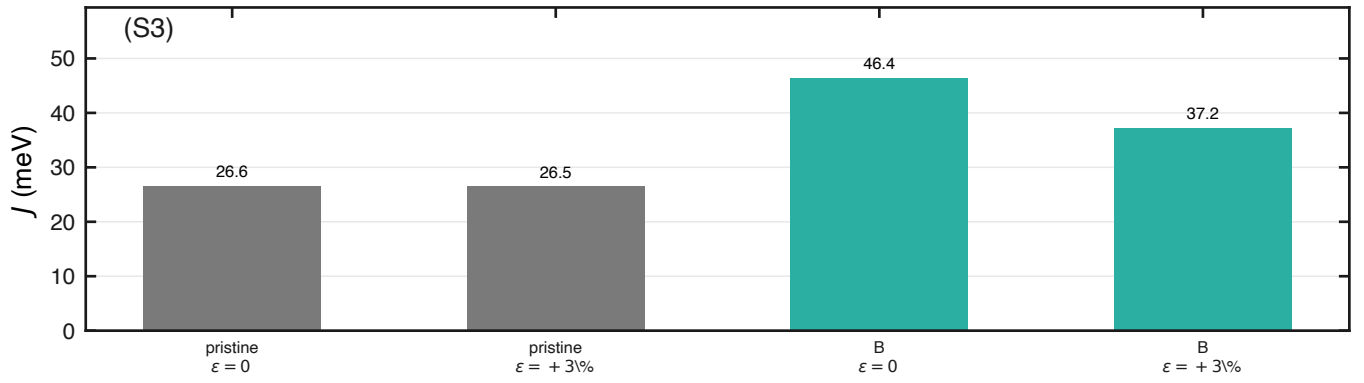


FIG. 5. Nearest-neighbor electronic coupling J on the tetramer dimer grid (pristine and B-doped cells at $\epsilon=0$ and $+3\%$; PBE+D3). Companion inverse-participation-ratio values for the same four corners are tabulated in the open data release (Methods).